



Building Multimodal Agents With NVIDIA Nemotron and RAPIDS - Part I

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Agenda

- Intro to Multimodality
- Image Embeddings
- Contrastive Language-Image Pretraining (CLIP)
- Generative Multimodality with Nemotron 3 Omni

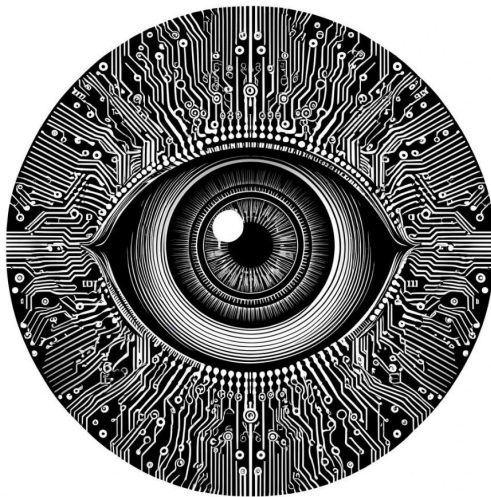
Notebooks

- Using Nemotron models through Nvidia Inference Microservices
- Screen recording understanding with Nemotron 3 Nano Omni
- Using Nemotron 3 Omni + Nemotron Parse 1.2 for document understanding
- A Nemotron + LangGraph agent for word counting

Setup

- [Configuring your NVIDIA API KEY to use models from build.nvidia.com](#)
- [Setting up a HuggingFace token](#)

Multimodal Agents

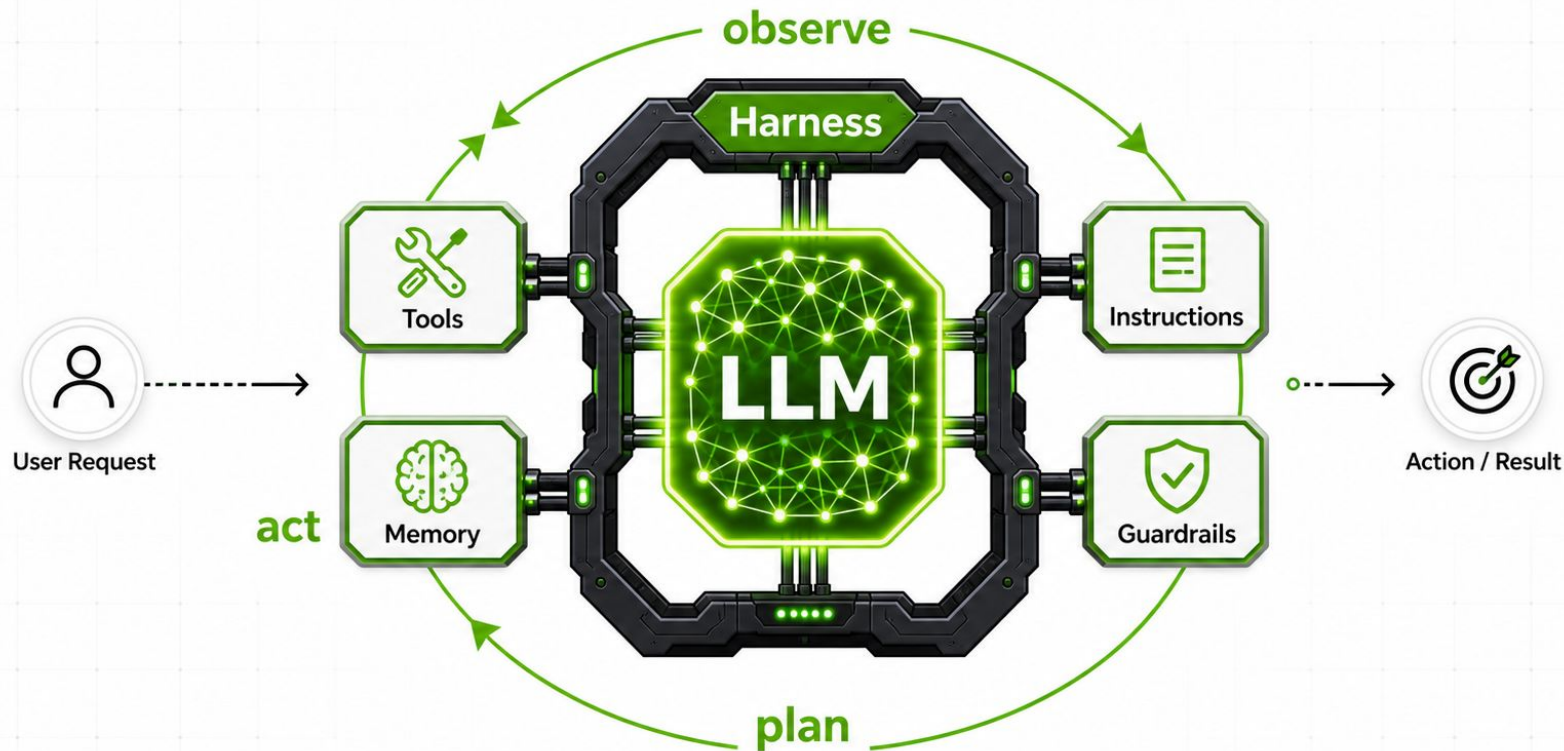


Text Generation with Large Language Models

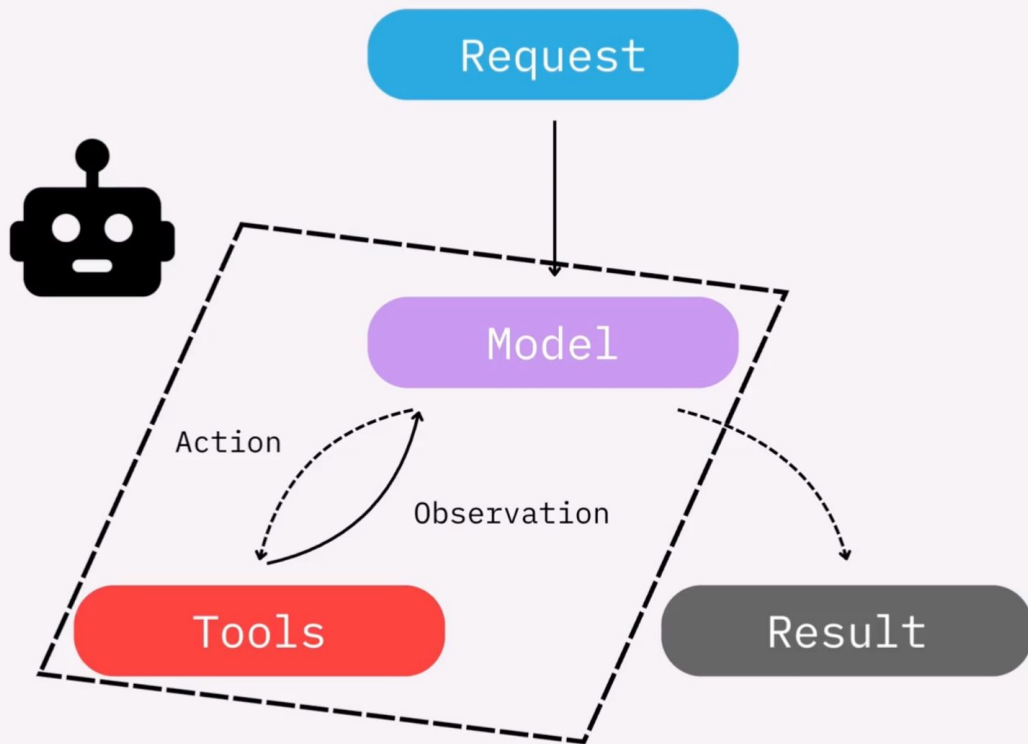


[Deep Dive into LLMs like ChatGPT](#)

AI Agent = LLM + Harness



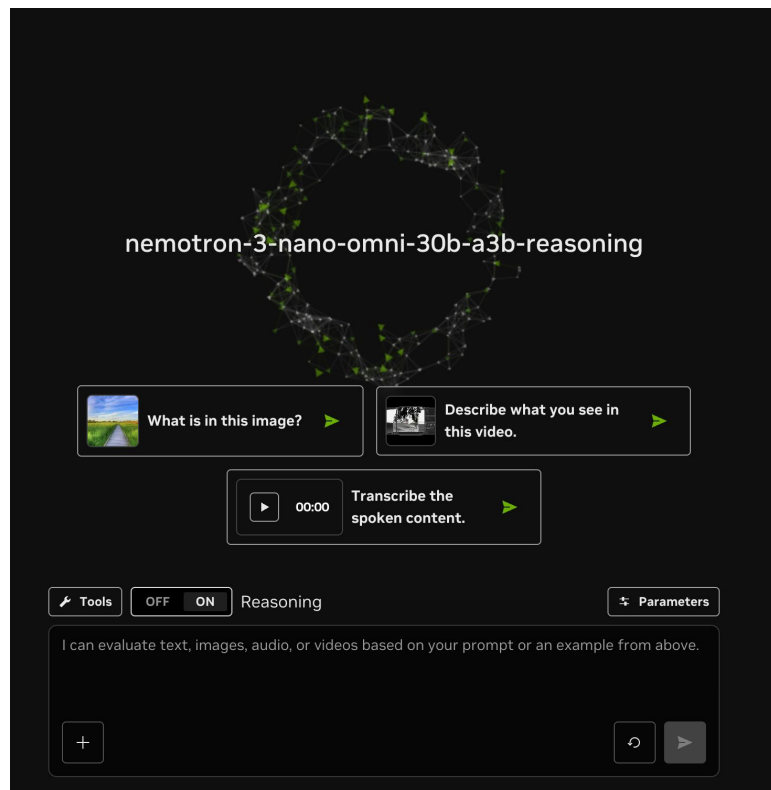
Reasoning LLMs as the central component of ReAct agents



[ReAct: Synergizing Reasoning and Acting in Language Models](#)

Nemotron 3 Nano Omni

Reasoning VLM trained on tool calling

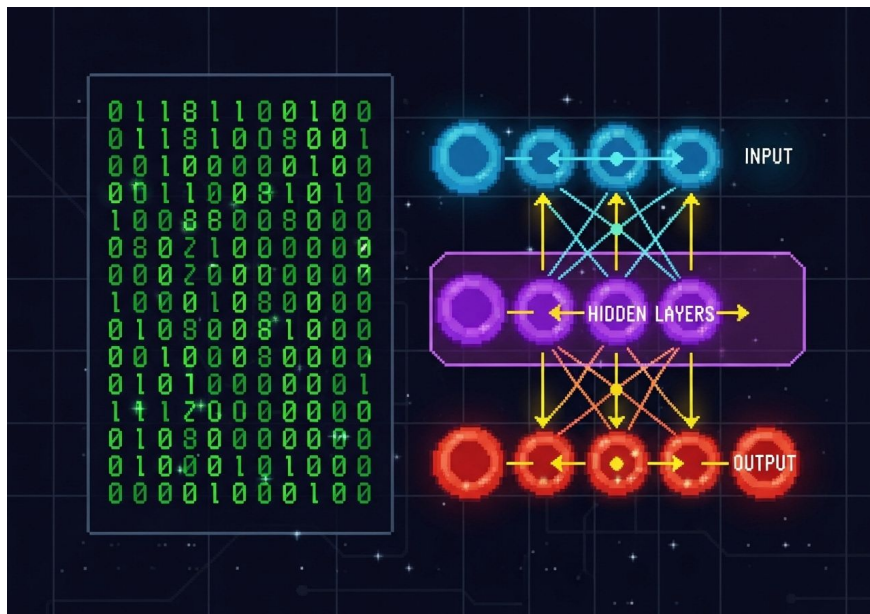


[nemotron-3-nano-omni-30b-a3b-reasoning](#) (live demo)

[Nemotron 3 Nano Omni: Efficient and Open Multimodal Intelligence](#) (ArXiv)

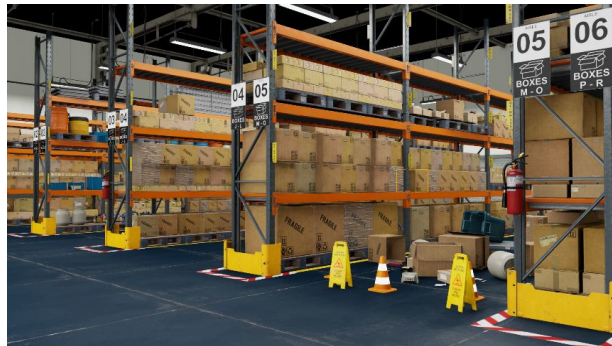
Working Hypothesis

Any data source that can be turned into a matrix can be used to train and run inference on a neural network

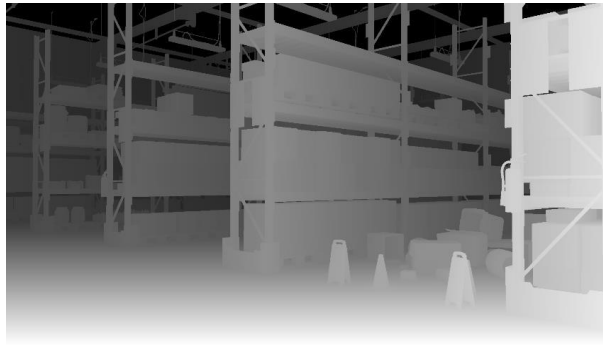


Sight

How A Computer Might See



RGB



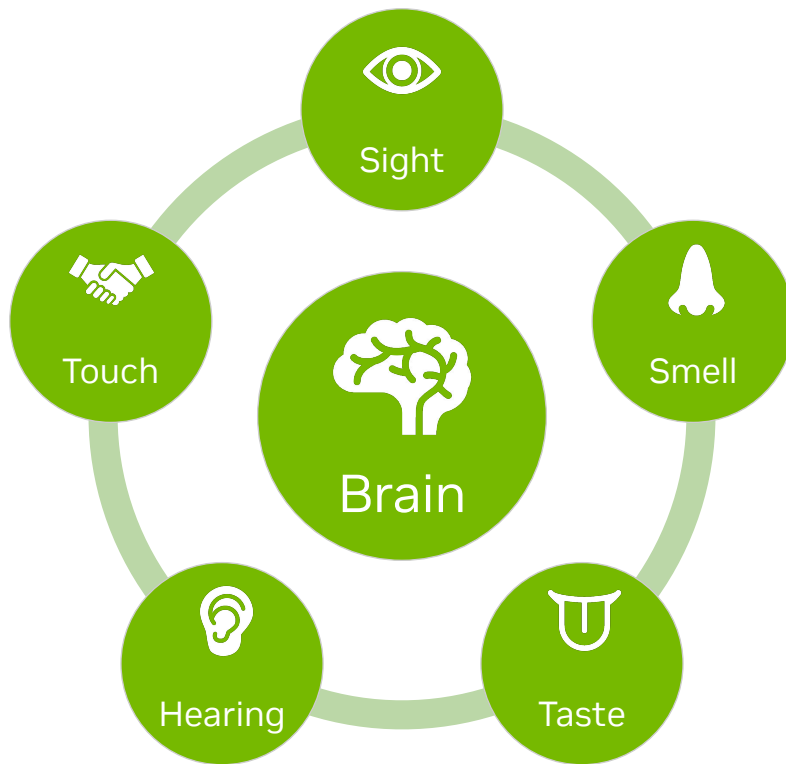
Depth



Fisheye

The Five Senses

How Humans Perceive



Smell and Taste

How A Computer Might Smell and Taste

pH



Nutrition Facts

Nutrition Facts	
1 serving per container	
Serving size	140 g
Amount per serving	
Calories	100
% Daily Value *	
Total Fat 0.5g	1%
Saturated Fat 0g	0%
Trans Fat 0g	
Cholesterol 0mg	0%
Sodium 0mg	0%
Total Carbohydrate 24g	9%
Dietary Fiber 6g	22%
Total Sugars 15g	
Includes 0g Added Sugars	0%
Protein <1g	
Vitamin D 0mcg	0%
Calcium 0mg	0%
Iron 0.3mg	2%
Potassium 120mg	2%
* The % Daily Value (DV) tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.	

Molecular Weight



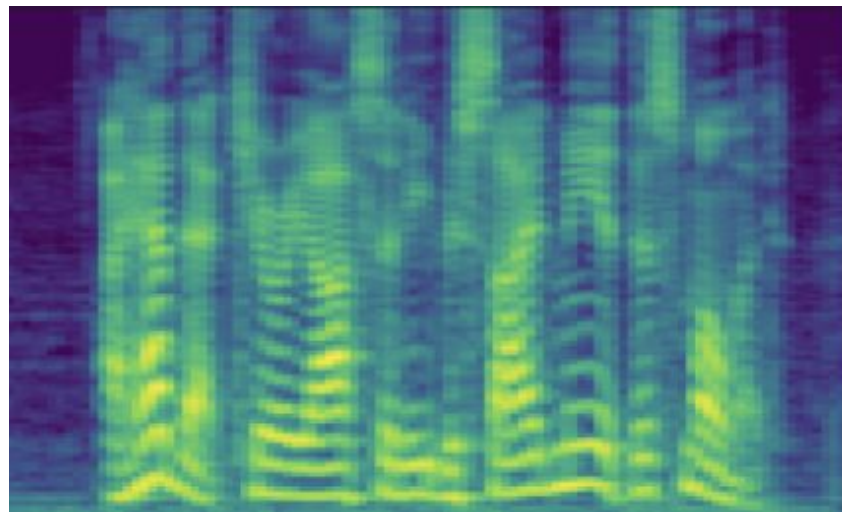
Hearing

How A Computer Might Hear

Sheet Music



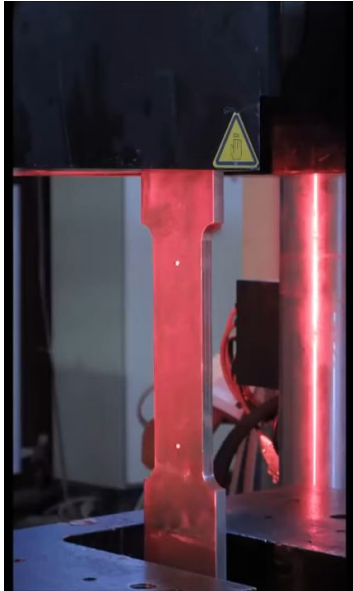
Spectrogram



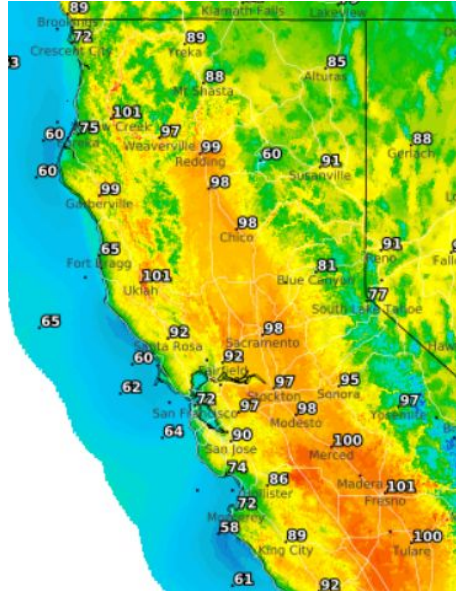
Touch

How A Computer Might Feel

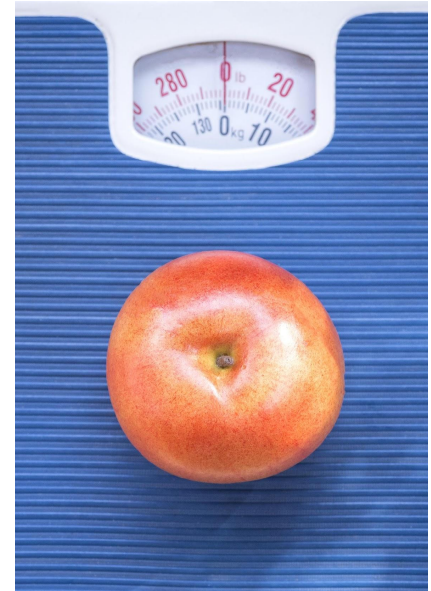
Tensile Strength



Weather Maps



Weight





Would you call this “multi-modal”?

Nemotron Parse

Creates additional context for Nemotron 3 Omni

nvidia

nemotron-parse [Downloadable](#)

Cutting-edge vision-language model excelling in retrieving text and metadata from images.

[document parsing](#) [supported language - english](#) [text and table extraction](#)

[Get API Key](#)

[Experience](#) [Model Card](#) [Deploy](#)

[API Reference](#) [Accelerated by DGX Cloud](#)

AI models generate responses and outputs based on complex algorithms and machine learning techniques, and those responses or outputs may be inaccurate, harmful,...

Input [Try](#) [Python](#) [Shell](#)

Upload Image [View Examples](#)

CUDA C++ Programming Guide, Release 12.8

Thread Scope	Description
cuda::thread_scope::thread_scope_thread	Only the CUDA thread which initiated asynchronous operations synchronizes.
cuda::thread_scope::thread_scope_block	All or any CUDA threads in the same GPU block as the initiating thread synchronizes.
cuda::thread_scope::thread_scope_device	All or any CUDA threads in the same GPU device as the initiating thread synchronizes.
cuda::thread_scope::thread_scope_system	All or any CUDA or CPU threads in the same system as the initiating thread synchronizes.

These thread scopes are implemented as extensions to standard C++ in the [CUDA Standard C++ library](#).

5.6. Compute Capability

The compute capability of a device is represented by a version number (also sometimes called the "SM version"). The version number identifies the features supported by the GPU hardware and is used for determining if certain instructions or CUDA features before and/or after certain are available on the device or not.

The compute capability comprises a major version number *M* and a minor version number *m* and is denoted as *CC.M.m*.

Devices with the same major version number are of the same architecture. The major version number is the device based on the CUDA major release. The device based on the hardware architecture is denoted as *CC.M*. The device based on the hardware architecture is denoted as *CC.M.m*.

The hardware architecture is denoted as *CC.M.m*.

GOVERNING TERMS: Access to this model is governed by the [NVIDIA API Trial Terms of Service](#); use of the model is governed by the [NVIDIA Community Model License](#). Use of the tokenizer included in this model is governed by the [CC-BY-4.0 license](#).

[Reset](#) [Run](#)

Output [Preview](#) [JSON](#)

Detected Elements

Page-header

CUDA C++ Programming Guide, Release 12.8

Table

Thread Scope	Description
cuda::thread_scope::thread_scope_thread	Only the CUDA thread which initiated asynchronous operations synchronizes.
cuda::thread_scope::thread_scope_block	All or any CUDA threads within the same thread block as the initiating thread synchronizes.
cuda::thread_scope::thread_scope_device	All or any CUDA threads in the same GPU device as the initiating thread synchronizes.
cuda::thread_scope::thread_scope_system	All or any CUDA or CPU threads in the same system as the initiating thread synchronizes.

Text

These thread scopes are implemented as extensions to standard C++ in the [CUDA Standard C++ library](#).

Section-header

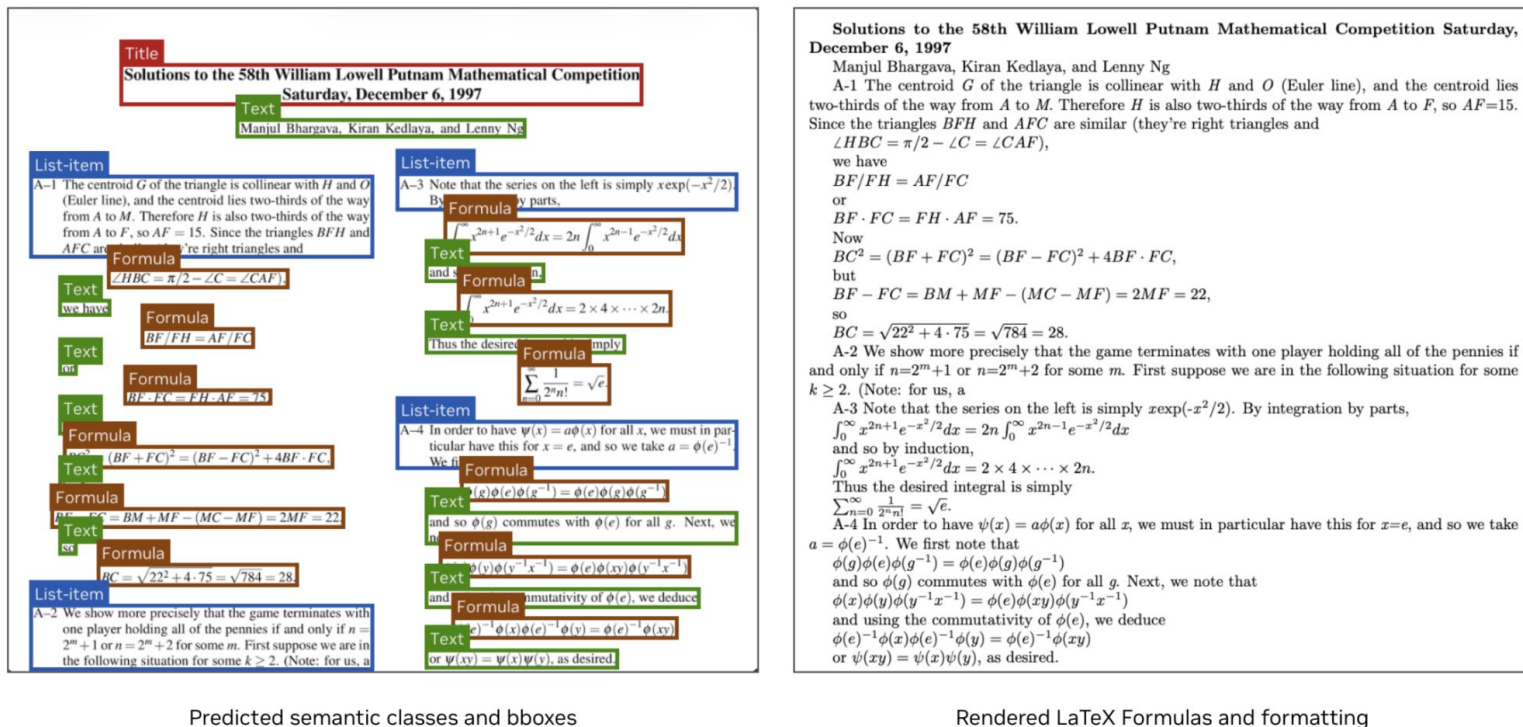
5.6. Compute Capability

Text

The compute capability of a device is represented by a version number (also sometimes called the "SM version").

<https://build.nvidia.com/nvidia/nemotron-parse>

 NVIDIA



Predicted semantic classes and bboxes

Rendered LaTeX Formulas and formatting

Figure 2 | OCR, extraction of text formatting and mathematical equations in LaTeX and markdown.

2.2.2. Output format

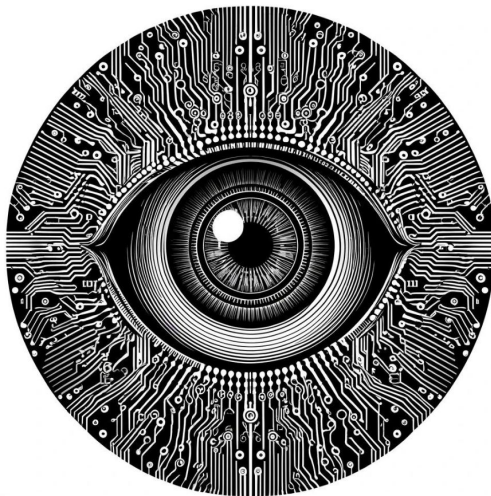
Nemotron-Parse predicts bounding boxes of the semantic blocks in the form of relative coordinates, in a scale of 1024×1280 . These bounding boxes are predicted in a canonical reading order, that includes Page-Header elements at the start of the page, followed by Text, Section-Header, List-Item, Title and Formula elements in the order as they would be read by a person looking at the given page, followed by Footnotes, Page-Footers, Tables, Pictures, and Captions in the end. Nemotron-Parse-TC improves upon this canonical reading order, and also includes non-reading-order (floating) elements, i.e., Footnotes, Page-Footers, Tables, Pictures, and Captions within the natural ordering of the page.

In the maximal information setting the output is in the following format:

```
<x_(\d+)><y_(\d+)>(.*?)<x_(\d+)><y_(\d+)>  
<class_(\w+)>
```

where the **first group** denotes the coordinates of the top-left corner, the **second group** denotes the text contained within the bounding box, the **third group** denotes the coordinates of the bottom-right corner, and the **final group** represents the semantic class. For example, `<x_0.1152><y_0.2586># NVIDIA Nemotron-Parse 1.1<x_0.8799><y_0.2797> <class_Title>`.

Image Embeddings



Learning goals

- Understand dense vector embeddings as learned representations
- Describe how image embeddings are produced in neural networks
- Extract image embeddings out of a pretrained Resnet50
- Inspect image neighborhoods using the cosine similarity metric

Sparse vs dense vector representations

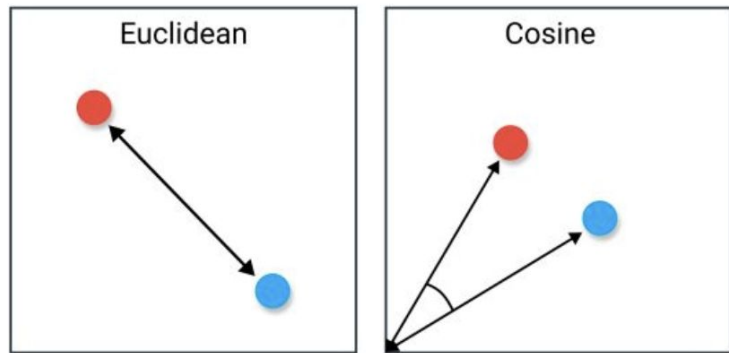
$$\mathbf{x}_1 = \text{[Image of handwritten 3]} \quad \mathbf{x}_2 = \text{[Image of handwritten 3]} \quad 28 \times 28 \text{ pixels} = 784 \text{ values}$$

$$\mathbf{x}_1 \text{ sparse} = \mathbf{x}_2 \text{ sparse} = [0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0]^\top$$

$$\mathbf{x}_2 \text{ dense} = [0.10 \quad -0.75 \quad 0.35 \quad 0.41 \quad -0.20 \quad 0.89 \quad -0.60 \quad 0.03 \quad 0.59 \quad -0.50]^\top$$

$$\mathbf{x}_1 \text{ dense} = [0.12 \quad -0.85 \quad 0.37 \quad 0.44 \quad -0.22 \quad 0.90 \quad -0.63 \quad 0.01 \quad 0.58 \quad -0.49]^\top$$

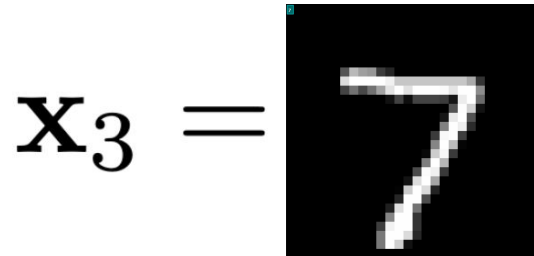
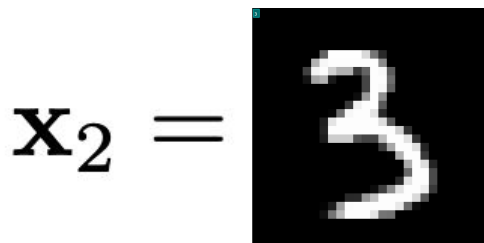
Common metrics to compare embeddings



$$d_{\text{euclid}}(x_1, x_2) = \sqrt{\sum_{i=1}^n (x_{1i} - x_{2i})^2}$$

$$\text{cosine similarity}(x_1, x_2) = \frac{x_1 \cdot x_2}{\|x_1\| \|x_2\|}$$

Similar objects have similar embeddings



Similar digits (X_1, X_2): High cos similarity (≈ 0.98), Low Euclidean distance (≈ 0.3)

Different digits (X_1, X_3): Low cos similarity (≈ 0.4), High Euclidean distance (≈ 2.1)

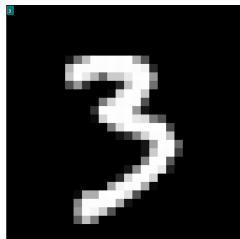
Embeddings as representations learned during training

```
import torch.nn as nn
```

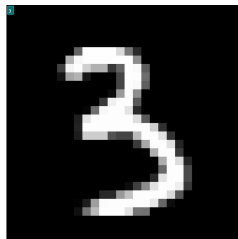
```
conv_linear_embedder = nn.Sequential(  
    nn.Conv2d(1, 32, kernel_size=3), # 26x26x32  
    nn.ReLU(),  
    nn.Conv2d(32, 64, kernel_size=3), # 24x24x64  
    nn.ReLU(),  
    nn.Flatten(), # 24*24*64  
    # This layer can be extracted as 64-dimensional  
    # embedding  
    nn.Linear(24*24*64, 64),  
    nn.ReLU(),  
    # This layer can be a 10-dimensional embedding  
    nn.Linear(in_features=64, out_features=10)  
)
```

$$H(p, q) = - \sum_i p(i) \log q(i)$$

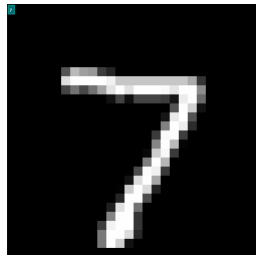
$\mathbf{x}_1 =$



$\mathbf{x}_2 =$

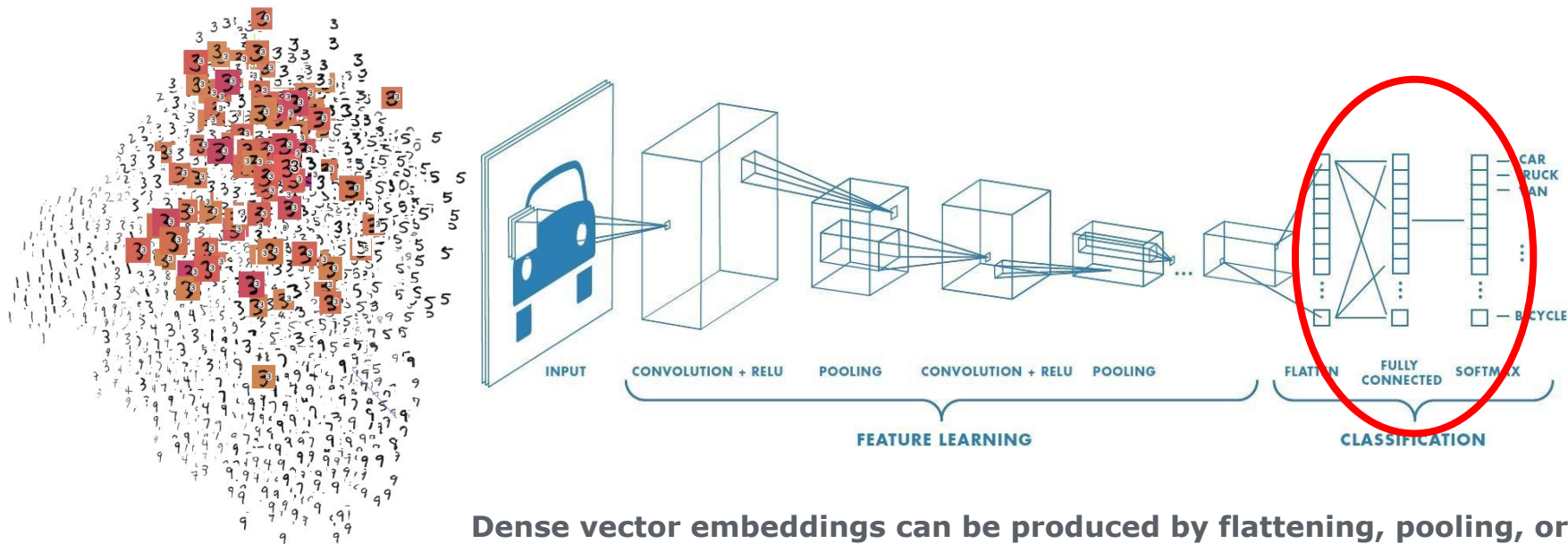


$\mathbf{x}_3 =$



Embeddings as a byproduct of training

When we train a neural network, we make it **embed** inputs that are semantically similar, close to each other



Dense vector embeddings can be produced by flattening, pooling, or feeding (learned) features to linear layers

[Source of image](#) (try exploring interactively)

Extract embeddings from a pretrained Resnet50

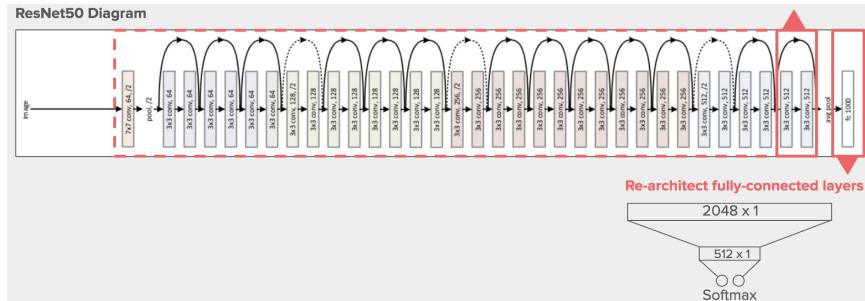
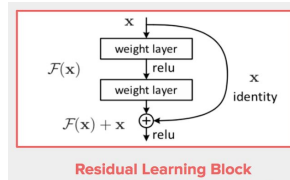
```
import torch
from torchvision import models
```

```
# Load pre-trained ResNet50 weights and transforms
weights = models.ResNet50_Weights.IMAGENET1K_V2
model = models.resnet50(weights=weights)
# Preprocess the image using the transformations from the imagenet dataset
preprocess = weights.transforms()
```

```
# Remove final Imagenet classification layer, keep 2048 dense vector
# Take time to unpack this syntax in the practical notebook
embedder = torch.nn.Sequential(*list(model.children())[:-1])
```

```
# Disable BatchNormalization
embedder.eval()
# Disable gradient computation
with torch.inference_mode():
```

```
    # unsqueeze(0) adds the batch size for inference
    # squeeze() removes the singleton dimensions from the output vector
    embedding = embedder(preprocess(image).unsqueeze(0)).squeeze()
```



Approaches to produce embeddings with neural networks

```
import torch.nn as nn
```

```
# Approach 1: Direct embedding
```

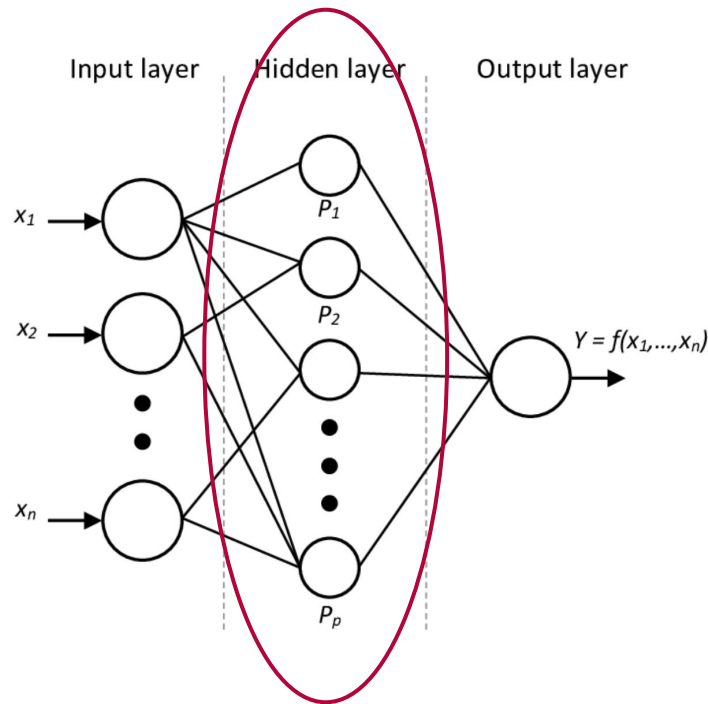
```
linear_embedder = nn.Sequential(  
    nn.Flatten(), # 784  
    nn.Linear(784, 10)  
)
```

```
# Approach 2: Convolution and then fully connected layer
```

```
conv_linear_embedder = nn.Sequential(  
    nn.Conv2d(1, 32, kernel_size=3), # 26x26x32  
    nn.ReLU(),  
    nn.Conv2d(32, 64, kernel_size=3), # 24x24x64  
    nn.ReLU(),  
    nn.Flatten(), # 24*24*64  
    nn.Linear(24*24*64, 10)  
)
```

```
# Approach 3: Global pooling after convolutions
```

```
conv_pool_embedder = nn.Sequential( # MNIST input is size 28x28  
    nn.Conv2d(1, 32, kernel_size=3), # 26x26x32  
    nn.ReLU(),  
    nn.Conv2d(32, 10, kernel_size=3), # 24x24x10  
    nn.ReLU(),  
    nn.AdaptiveAvgPool2d((1, 1)), # 1x1x10  
    nn.Flatten() # 10  
)
```



$$L_{L1} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

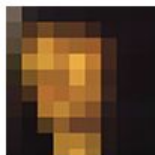
README.md

Art Recommendation system

This is the repository of a portfolio project at [DSR](#). This project aims to identify similar images using pre-trained computer vision networks. For an explanation of the technology see the [technology section](#).

Contributors

- [Catarina Ferreira](#)
- [Gargi Maheshwari](#)

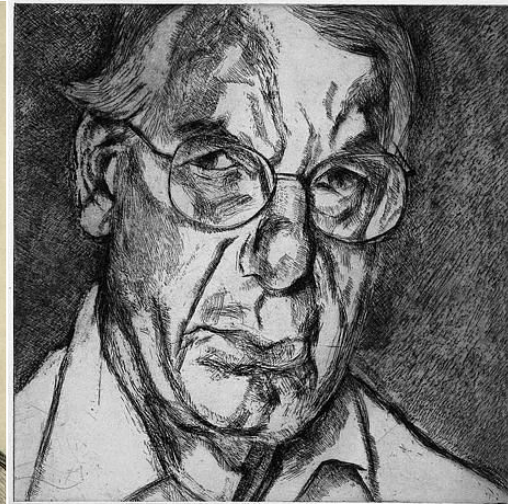


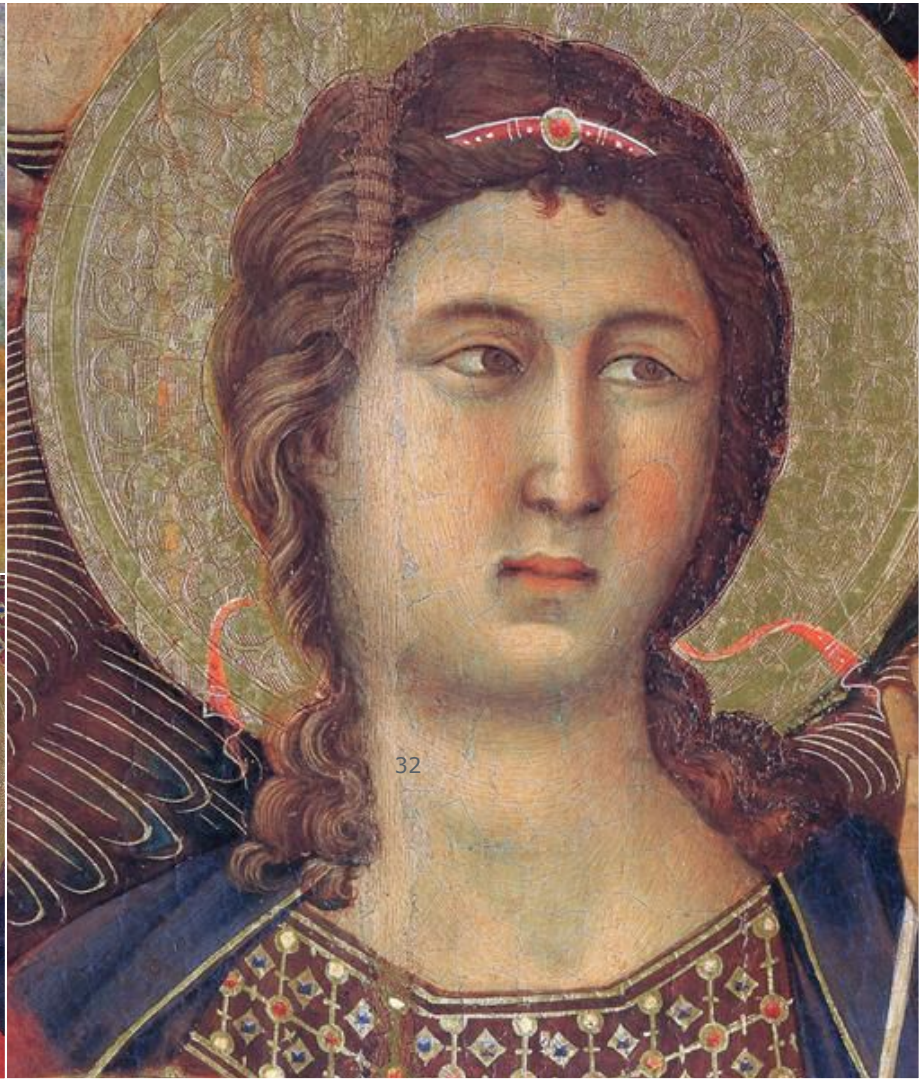
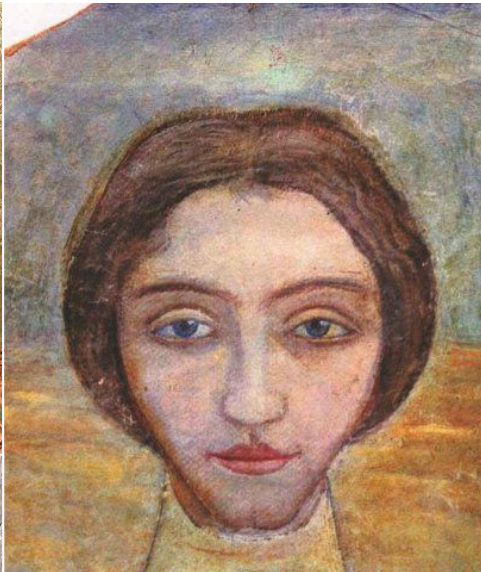
WIKIART
VISUAL ART ENCYCLOPEDIA

<https://github.com/gargimaheshwari/Wikiart-similar-art>





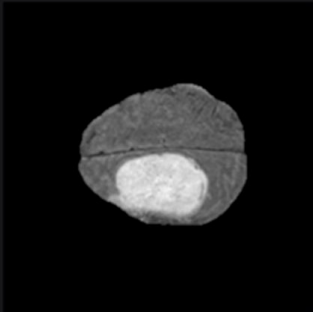




Application: differential diagnosis

PIXEL DIAGNOSE

Select Image



Cancer Type:

- ☒ Gliomas
- ☒ Meningiomas
- ☒ Metastasis

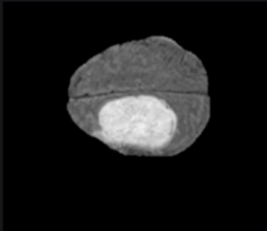
Image Type:

- ☐ t1
- ☐ t1c
- ☐ t2
- ☒ flair
- ☐ seg_t1
- ☐ seg_t1c
- ☐ seg_t2
- ☐ seg_flair

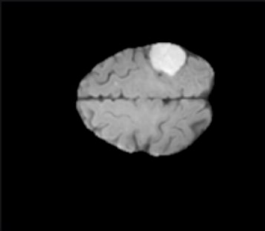
Number of Results per Type:

9

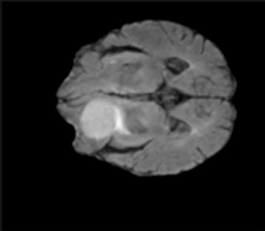
Start Search



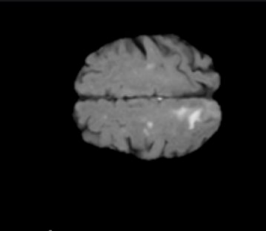
Slice no.: 112 / 150
Similarity Score: 0.9999999



Slice no.: 114 / 150
Similarity Score: 0.8529618



Slice no.: 62 / 150
Similarity Score: 0.82390577



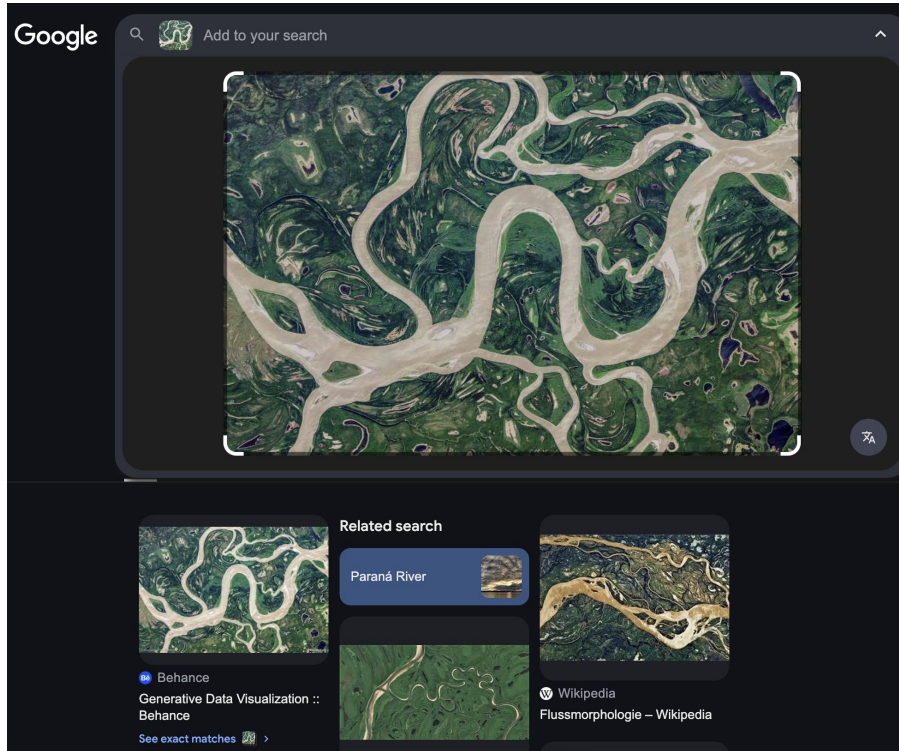
Slice no.: 109 / 150
Similarity Score: 0.8092099



Slice no.: 119 / 150
Similarity Score: 0.8026103

Image from <https://pixel-diagnose.github.io/>

Application: search by image



Search for similar images in images.google.com

Summary

Embeddings are learned vector representations

- Neural networks learn to map similar images to similar vectors through training
- We measure the similarity of embeddings using metrics like cosine similarity and Euclidean distance

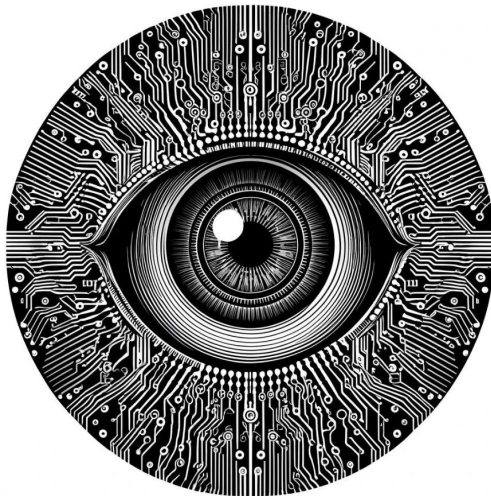
There are different architectural approaches to create embeddings

- We can use linear layers, or flattened and pooled convolutions to produce embeddings
- Imagenet-pretrained models produce rich representations in their embeddings

Embeddings have multiple applications in semantic search and dataset curation

- They can be used for reverse image search, recommendation systems, and deduplication

Contrastive Language-Image Pretraining (CLIP)



Learning goals

- Understand how CLIP models are trained
- Classify images with CLIP using a zero-shot approach
- Create image and text embeddings with a pretrained CLIP model
- Discuss CLIP's uses and limitations

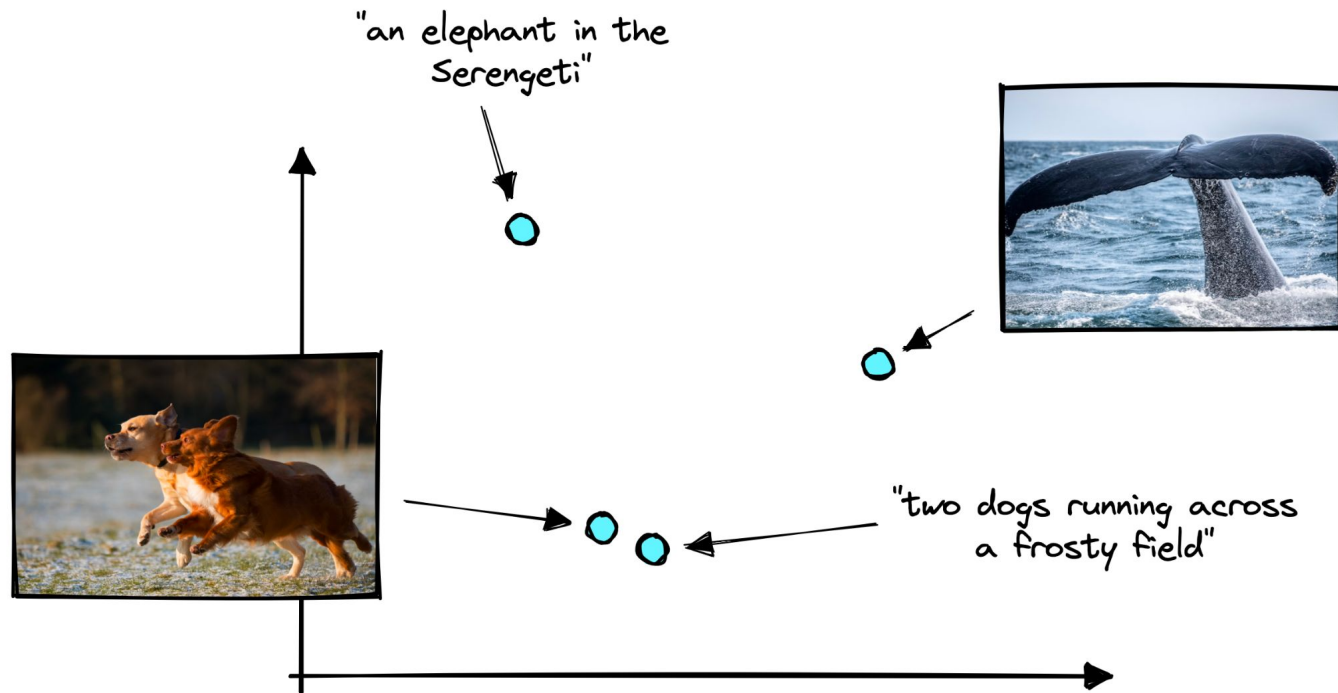
CLIP: ‘Contrastive Language Image Pretraining’

Learning Transferable Visual Models From Natural Language Supervision

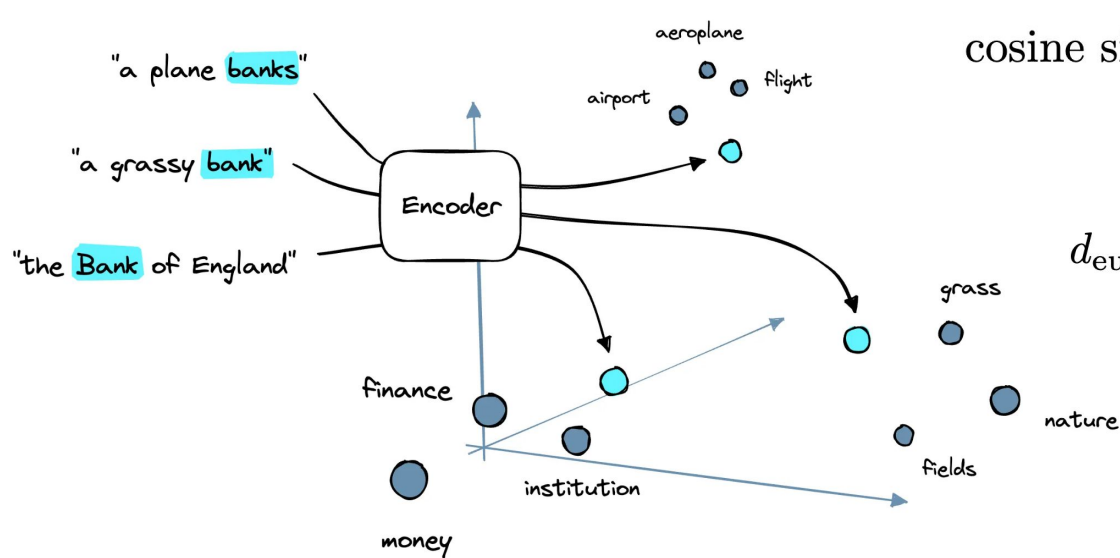
Alec Radford^{*1} Jong Wook Kim^{*1} Chris Hallacy¹ Aditya Ramesh¹ Gabriel Goh¹ Sandhini Agarwal¹
Girish Sastry¹ Amanda Askell¹ Pamela Mishkin¹ Jack Clark¹ Gretchen Krueger¹ Ilya Sutskever¹

- Connects text and images in a shared embedding space
- Created by OpenAI in 2021
- Trained on **400M image-textual description pairs** scraped from the internet
- Predicts the most relevant text snippet given an image, enabling “zero-shot classification”

Aligning text and image embeddings



Text encoders

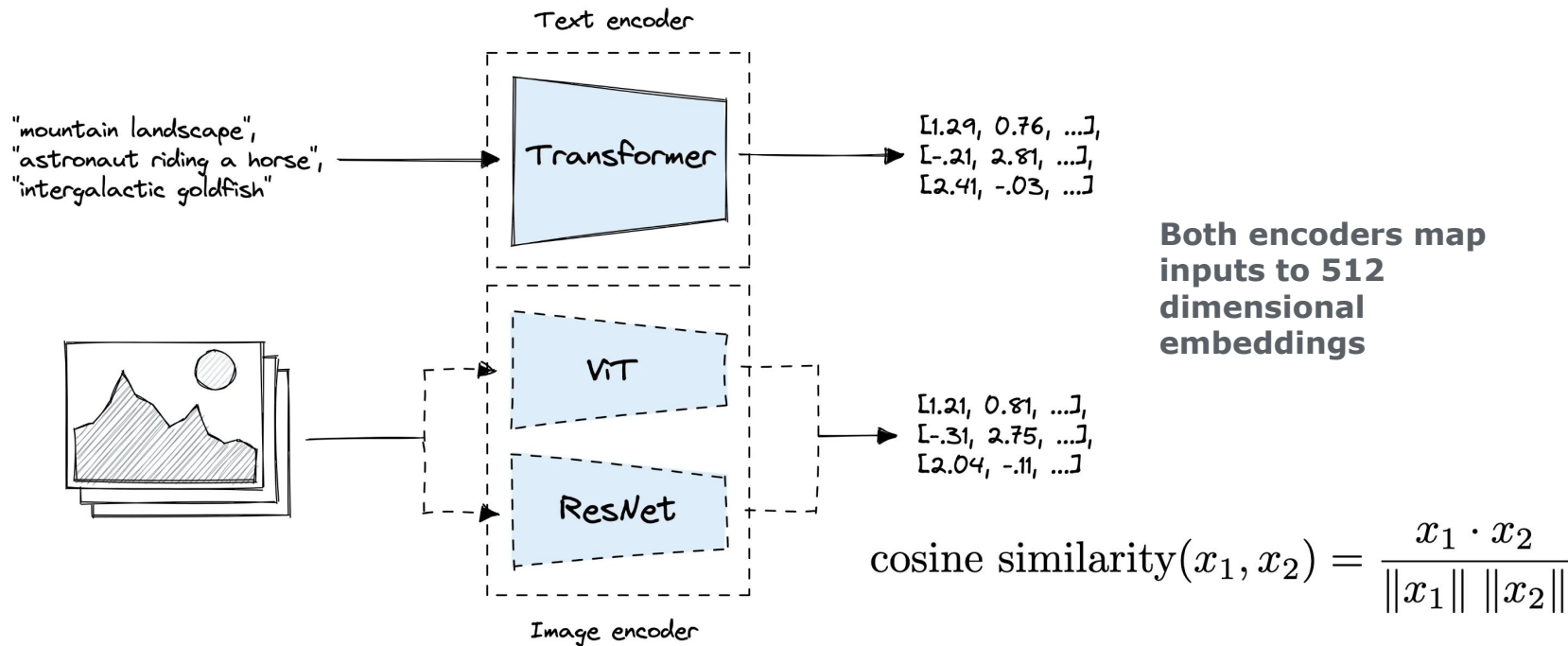


$$\text{cosine similarity}(x_1, x_2) = \frac{x_1 \cdot x_2}{\|x_1\| \|x_2\|}$$

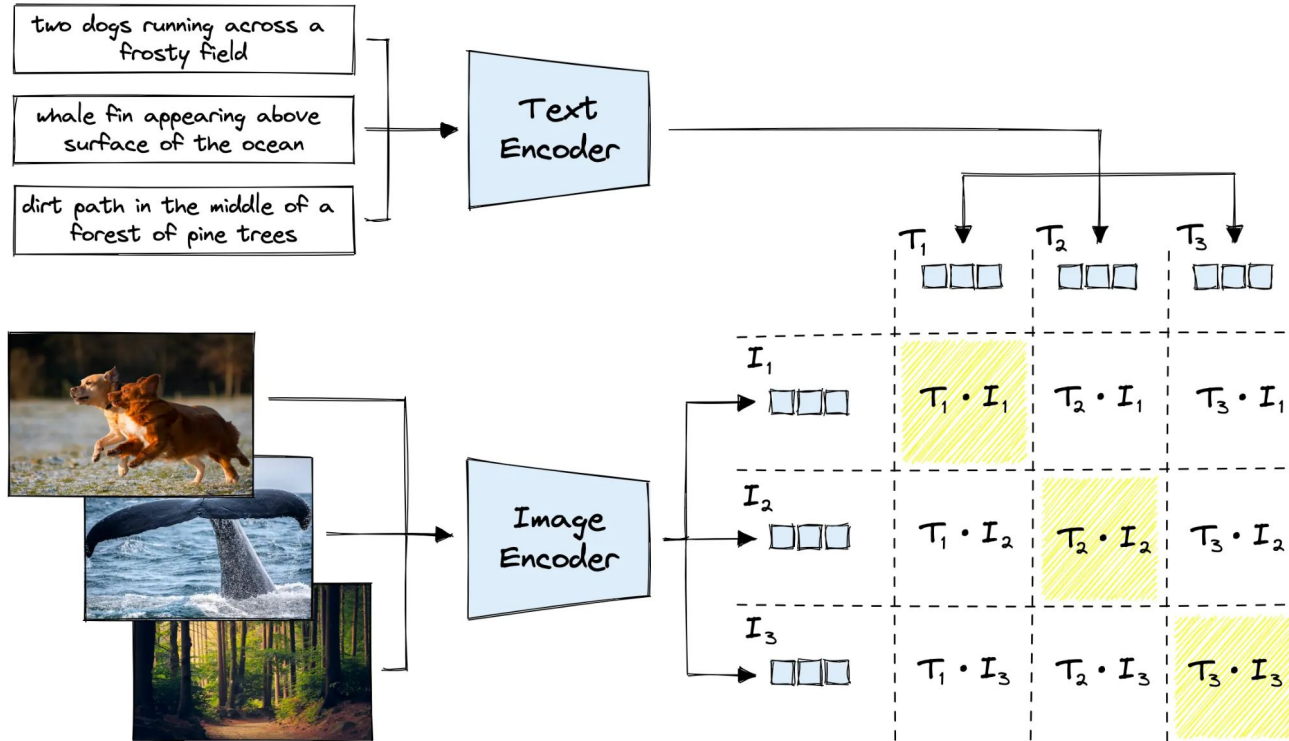
$$d_{\text{euclid}}(x_1, x_2) = \sqrt{\sum_{i=1}^n (x_{1i} - x_{2i})^2}$$

Image from <https://www.pinecone.io/learn/series/image-search/vision-transformers/>

CLIP's architecture



Maximizing cosine similarity of matching text and image embeddings



Training algorithm

```
# image_encoder - ResNet or Vision Transformer
# text_encoder - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l] - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T) #[n, d_t]

# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss = (loss_i + loss_t)/2
```

Figure 3. Numpy-like pseudocode for the core of an implementation of CLIP.

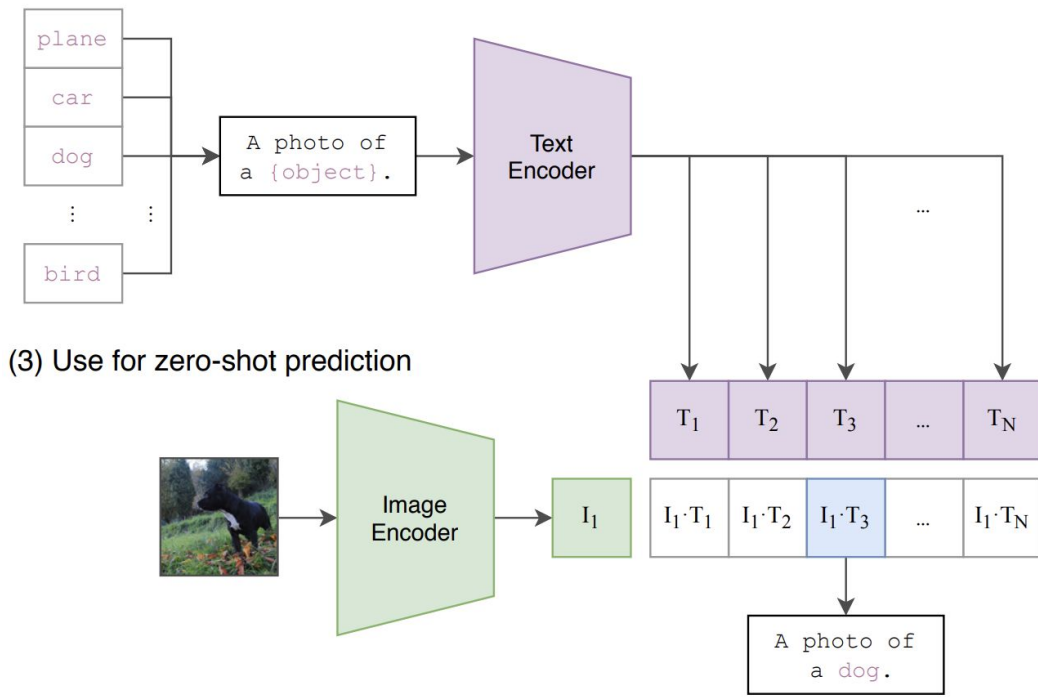
	T ₁	T ₂	T ₃	...	T _N
I ₁	I ₁ ·T ₁	I ₁ ·T ₂	I ₁ ·T ₃	...	I ₁ ·T _N
I ₂	I ₂ ·T ₁	I ₂ ·T ₂	I ₂ ·T ₃	...	I ₂ ·T _N
I ₃	I ₃ ·T ₁	I ₃ ·T ₂	I ₃ ·T ₃	...	I ₃ ·T _N
⋮	⋮	⋮	⋮	⋮	⋮
I _N	I _N ·T ₁	I _N ·T ₂	I _N ·T ₃	...	I _N ·T _N

$$H(p, q) = - \sum_i p(i) \log q(i)$$

$$H(q, p) = - \sum_i q(i) \log p(i)$$

$$\text{symmetric CE loss} = \frac{H(p, q) + H(q, p)}{2}$$

Zero-shot classification with CLIP



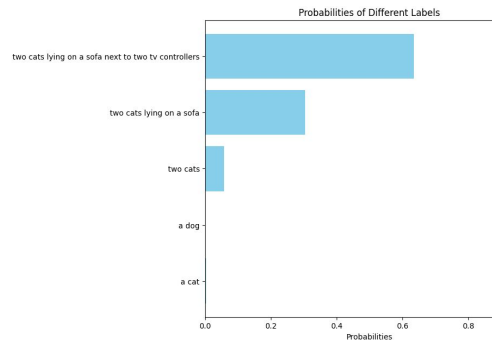
The model is able to create good classifiers without extra training

$$\text{softmax}(x_i, T) = \frac{e^{x_i/T}}{\sum_{j=1}^n e^{x_j/T}}$$

Producing embeddings with CLIP (1/2)

```
import torch
from transformers import CLIPProcessor, CLIPModel
from PIL import Image

# Load model and inputs
model =
CLIPProcessor.from_pretrained("openai/clip-vit-base-patch32")
processor =
CLIPModel.from_pretrained("openai/clip-vit-base-patch32")
image = Image.open("cats.jpg")
cat_text = ['a cat', 'a dog', 'two cats',
            'two cats lying on a sofa',
            """two cats lying on a sofa
            next to two tv controllers"""]
```



Producing embeddings with CLIP (2/2)

Get embeddings

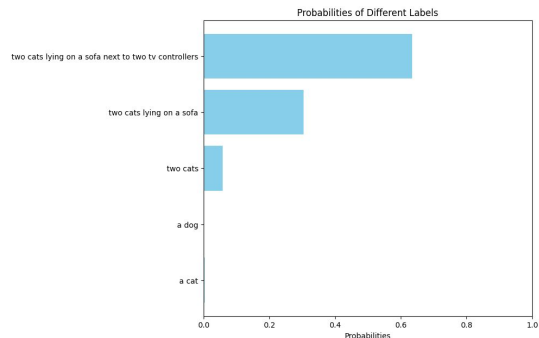
```
inputs = processor(text=cat_text, images=image, return_tensors="pt",  
padding=True)  
outputs = model(**inputs)
```

Calculate similarities

```
sims = torch.nn.functional.cosine_similarity(  
    outputs.image_embeds[:, None],  
    outputs.text_embeds[None, :],  
    dim=-1  
)
```

Print results

```
for text, sim in zip(cat_text, sims[0]):  
    print(f"{text}: {sim:.3f}")
```



Zero-shot classification with CLIP

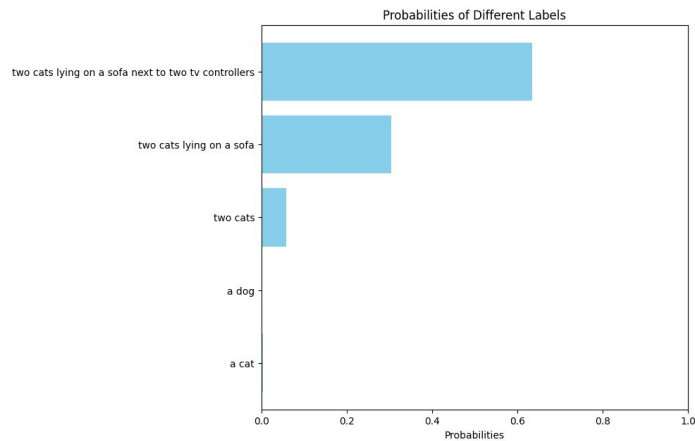
Notice what happens with the output probabilities

when we make the labels more specific

```
cat_text = ['a cat',  
            'a dog',  
            'two cats',  
            'two cats lying on a sofa',  
            'two cats lying on a sofa next to two tv controllers']
```

```
inputs = processor(text=cat_text,  
                  images=[cats_img],  
                  return_tensors="pt", padding=True)  
cat_outputs = model(**inputs)  
cat_logits_per_image = cat_outputs.logits_per_image  
cat_probs = (cat_logits_per_image/temperature).softmax(dim=1)
```

$$\text{softmax}(x_i, T) = \frac{e^{x_i/T}}{\sum_{j=1}^n e^{x_j/T}}$$



Transferable representations: CLIP against a ResNet101 pretrained on Imagenet

	Dataset Examples	ImageNet ResNet101	Zero-Shot CLIP	Δ Score
ImageNet		76.2	76.2	0%
ImageNetV2		64.3	70.1	+5.8%
ImageNet-R		37.7	88.9	+51.2%
ObjectNet		32.6	72.3	+39.7%
ImageNet Sketch		25.2	60.2	+35.0%
ImageNet-A		2.7	77.1	+74.4%

Training Resnet101: about 90 V100 GPU hours (4 days)
 Training CLIP ViT-L/14: about ~16,000 V100 GPU hours (666 days)

Note that CLIP was trained on a dataset of **400 million images** scraped from the Internet

The Imagenet1K dataset has only **1.28 million training images**

CLIP is a much more capable "foundation model"

Resnet101

~44.5 million parameters

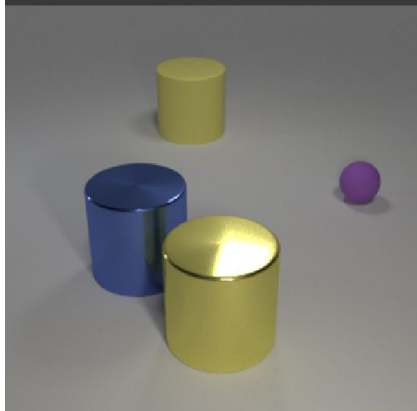
CLIP ViT-L/14@336px:

~428 million parameters

Limitations against fully supervised models

CLEVR Count

4 (75.0%) Ranked 2 out of 8 labels



× a photo of 3 objects.

✓ a photo of 4 objects.

× a photo of 5 objects.

× a photo of 6 objects.

× a photo of 10 objects.

Image from

<https://openai.com/index/clip/>

The authors comment on the limitations section of the paper that CLIP often fails to perform as well as a fully supervised model fine-tuned for the task

Notable examples:

- digit recognition (MNIST)
- tumor classification (PatchCamelyon)
- satellite imaging (EuroSAT)
- object counting (CLEVR Count)

CLIP guides image generation of diffusion models

Prompt:

"Create an image
of an astronaut
riding a horse in
pencil drawing
style"

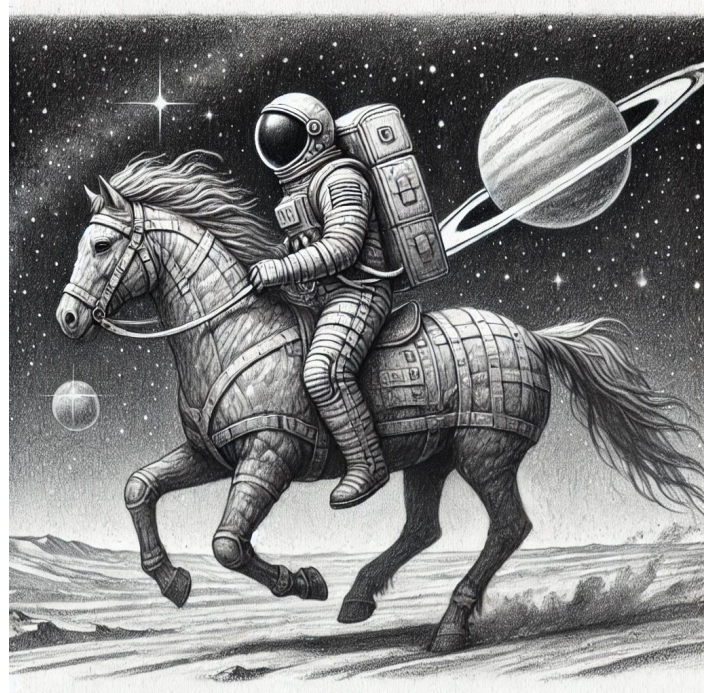
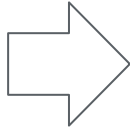


Image by OpenAI's Dall-E 3

Summary

CLIP learns joint embeddings

- CLIP creates a shared embedding space for images and text
- These multimodal embeddings have applications on semantic search and image generation

CLIP excels at zero-shot classification

- CLIP performs well on unseen tasks without additional training

CLIP has limitations

- CLIP is not always better than models fine-tuned for specific tasks
- Retraining CLIP with a ViT base is expensive both computationally and data-wise

Further reading

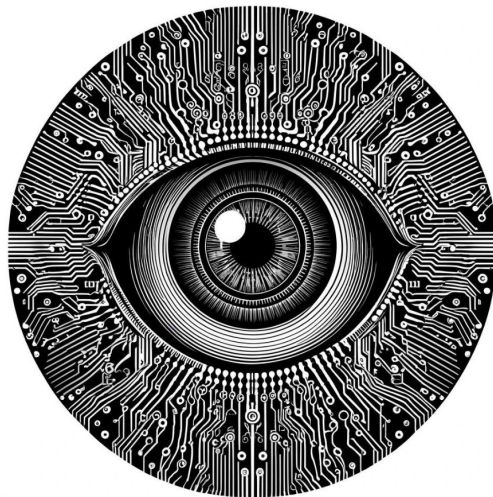
Learning Transferable Visual Models From Natural Language Supervision

- <https://arxiv.org/abs/2103.00020>

Multi-modal ML with OpenAI's CLIP

- <https://www.pinecone.io/learn/series/image-search/clip/>

Generative Multimodality with Nemotron 3 Omni



Learning goals

- Understand how combining embeddings create multimodal spaces
- Gain intuitions about the use of multimodal decoders for generation

nemotron-3-nano-omni-30b-a3b-reasoning



What is in this image? ➤



Describe what you see in this video. ➤



00:00

Transcribe the spoken content. ➤



Tools

OFF

ON

Reasoning



Parameters

I can evaluate text, images, audio, or videos based on your prompt or an example from above.

+



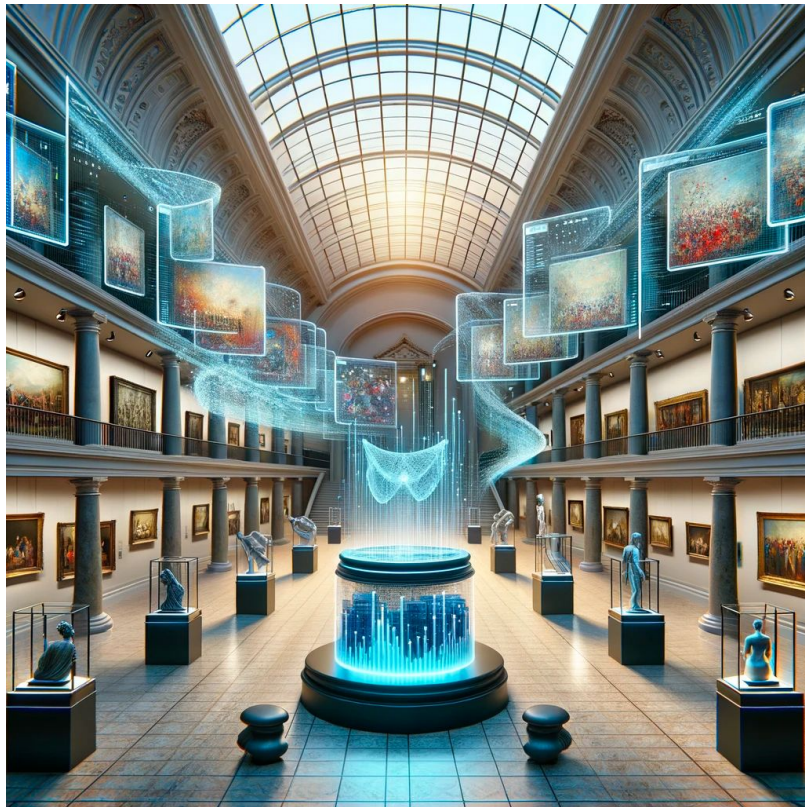
[Live demo at build.nvidia.com](https://build.nvidia.com)

[5 min video walkthrough](#)

Describing Multimodal Data with Nemotron 3 Omni



*"A galloping
horse, seen
from the side,
with the jockey
riding it"*



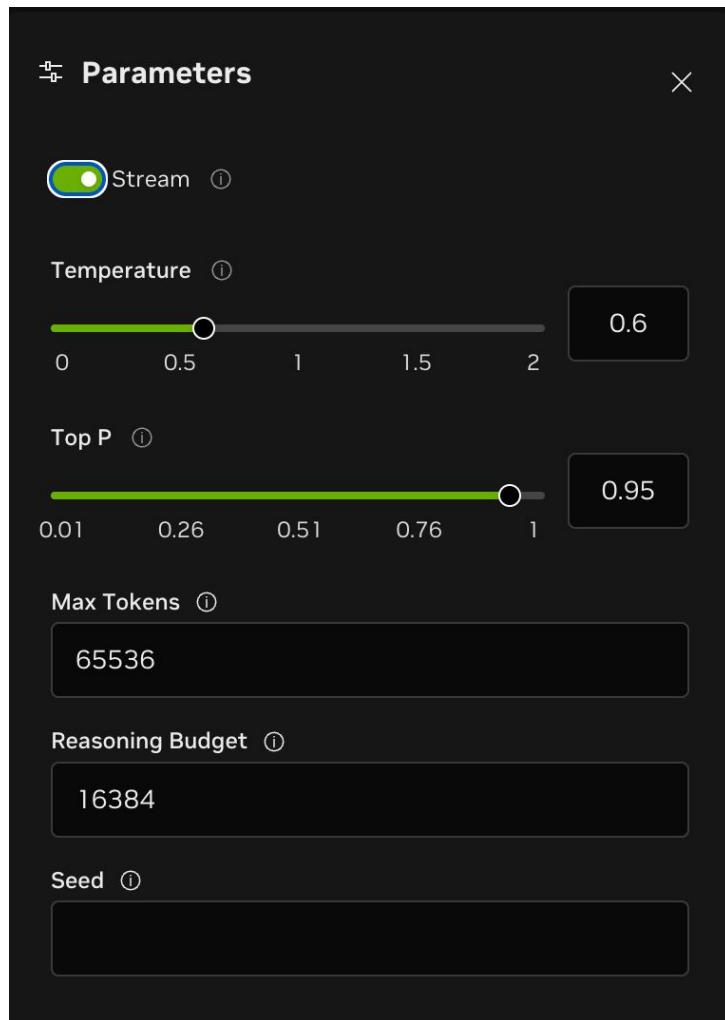
Use cases

- Video search and summarization 🎬
- Multimodal document understanding 📖
- Audio transcription 🗣️
- Dataset curation 🔧

Understand the architecture to understand its parameters



<https://build.nvidia.com/nvidia/nemotron-3-nano-omni-30b-a3b-reasoning>



4.5. Reasoning budget control

We study the effect of inference-time reasoning budgets by evaluating model performance under two settings: (1) a base configuration with a maximum sequence length of 16,384 tokens, and (2) a reasoning-enabled configuration with a 13K reasoning budget, a 1,024-token grace period, and a maximum sequence length of 16,384 tokens (Table 11).

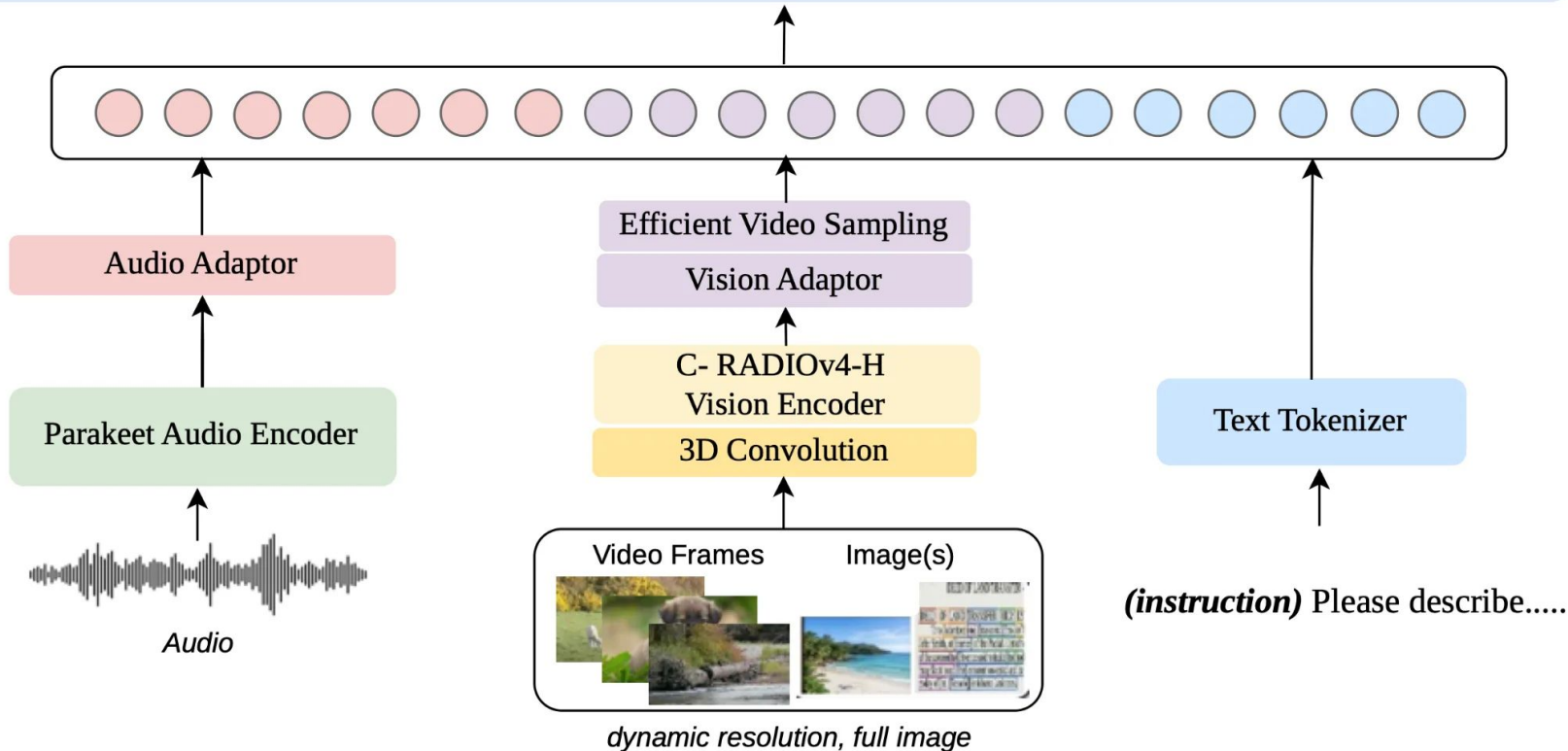
Our results suggest that reasoning budget adjustment yields accuracy gains on select benchmarks under reasoning-on mode, with no degradation observed on the remaining ones. These gains with budget control may arise from the early termination of malformed reasoning traces with repetition loops on out-of-distribution tasks, as well as the truncation of overly verbose reasoning chains for problems requiring minimal or straightforward reasoning.

Benchmark	MathVista-Mini	MMLongBench-Doc	DocVQA (Val)	Charxiv(RQ)	RefCOCO	VideoMME
w.o. reasoning budget	80.3	54.5	95.3	61.8	90.4	67.5
w. reasoning budget	82.8	56.8	95.2	64	90.6	70.3

Table 11 | Effect of reasoning budget across several key benchmarks.

There is a

Nemotron 3 Nano 30B-A3B LLM



NVIDIA's C-RADIO for Text ↔ Image

Text Prompt: "shoe"



Text Prompt: "helmet"



<https://huggingface.co/nvidia/C-RADIOv4-H>

[C-RADIOv4 \(Tech Report\)](#)

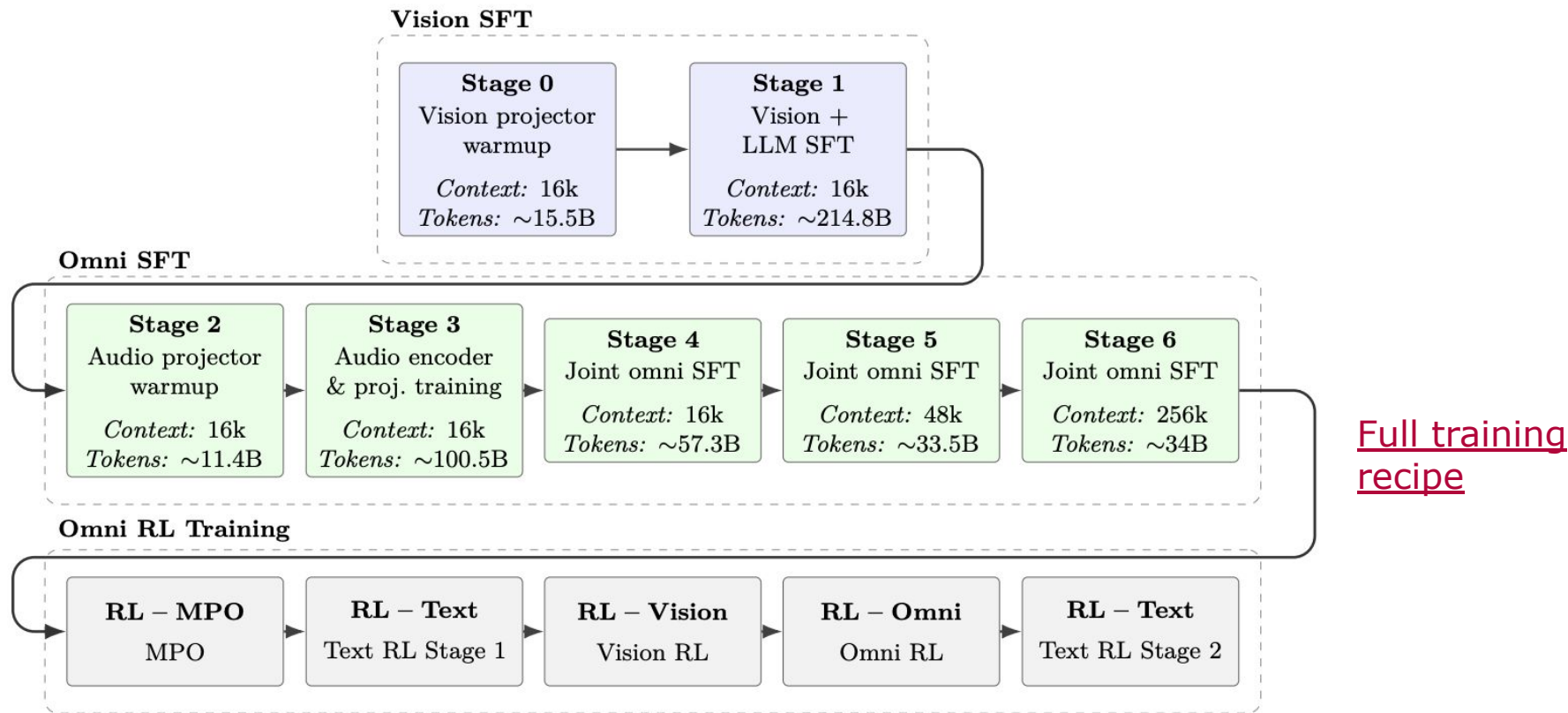


Figure 2 | Staged training recipe for the v3 omni-modal model. The pipeline first performs vision SFT, then joint omni SFT while progressively extending context length, followed by omni-modal RL training.

Long-document visual reasoning

Adobe 2015 10-K • 116-page filing



Handles questions over a 100+ page document

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549
FORM 10-K

(Mark One)
☒ ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d)
OF THE SECURITIES EXCHANGE ACT OF 1934
For the fiscal year ended November 27, 2015

☐ TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d)
OF THE SECURITIES EXCHANGE ACT OF 1934
For the transition period from _____ to _____
Commission File Number: 0-15175

Adobe
ADOBE SYSTEMS INCORPORATED
(Exact name of registrant as specified in charter)

Delaware
(State or other jurisdiction of incorporation or organization)

77-0818522
(U.S. Employer Identification No.)

345 Park Avenue, San Jose, California 95110-2794
(Address of principal executive office)

(408) 536-6000
(Registrant's telephone number, including area code)

Securities registered pursuant to Section 12(b) of the Act:

Title of Each Class	Name of Each Exchange on Which Registered
Common Stock, \$0.0001 par value per share	The NASDAQ Stock Market LLC (NASDAQ Global Select Market)

Securities registered pursuant to Section 12(g) of the Act: None

DOCUMENTS INCORPORATED BY REFERENCE
Portions of the Proxy Statement for the registrant's 2016 Annual Meeting of Stockholders (the "Proxy Statement"), to be filed within 120 days of the end of the fiscal year ended November 27, 2015, are incorporated by reference in Part III hereof. Except with respect to information specifically incorporated by reference in this Form 10-K, the Proxy Statement is not deemed to be filed as part hereof.



Scans until page 63 to find relevant info



Pages 1–62 scanned
Not containing the needed answer

First relevant evidence

63

ADOBE SYSTEMS INCORPORATED
CONSOLIDATED STATEMENTS OF CASH FLOWS
(In thousands)

	Years Ended		
	November 27, 2015	November 28, 2014	November 29, 2013
Cash flows from operating activities:			
Net cash provided by operating activities	\$ 1,469,502	\$ 1,287,482	\$ 1,351,684
Cash flows from investing activities:			
Net cash used for investing activities	(1,488,358)	(490,707)	(1,177,803)
Cash flows from financing activities:			
Net cash used for financing activities	(200,687)	(597,283)	(539,138)
Effect of foreign currency exchange rates on cash and cash equivalents	(21,297)	(6,448)	(5,243)
Net increase (decrease) in cash and cash equivalents	(240,840)	282,844	(360,496)
Cash and cash equivalents at beginning of year	1,117,400	834,556	1,425,052
Cash and cash equivalents at end of year	\$ 876,560	\$ 1,117,400	\$ 834,556

See accompanying Notes to Consolidated Financial Statements.

63

First relevant evidence

Second relevant evidence

99

ADOBE SYSTEMS INCORPORATED
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS (Continued)

NOTE 16. DEBT

Our debt as of November 27, 2015 and November 28, 2014 consisted of the following (in thousands):

	2015	2014
Notes	\$ 1,387,410	\$ 1,496,778
Fair value of interest rate swap	19,821	14,268
Adjusted carrying value of Notes	1,907,231	1,511,046
Capital lease obligations	—	3,269
Total debt and capital lease obligations	1,907,231	1,514,315
Less: current portion	—	603,229
Debt and capital lease obligations	\$ 1,907,231	\$ 911,086

99

Second relevant evidence

116

Page 116
End of filing

Distant pages
63 → 99
combine evidence
to answer

Title page (Page 1)

Question

What is Cash Flow to Total Debt Ratio for ADBE in FY2015?

Ground truth

0.77

Prediction

0.77



Reasoning summary

Page 63: FY2015 operating cash flow = \$1,469,502k

Page 99: FY2015 total debt = \$1,907,231k

Compute: $1,469,502 / 1,907,231 = 0.7705$

Final answer: 0.77



How it works



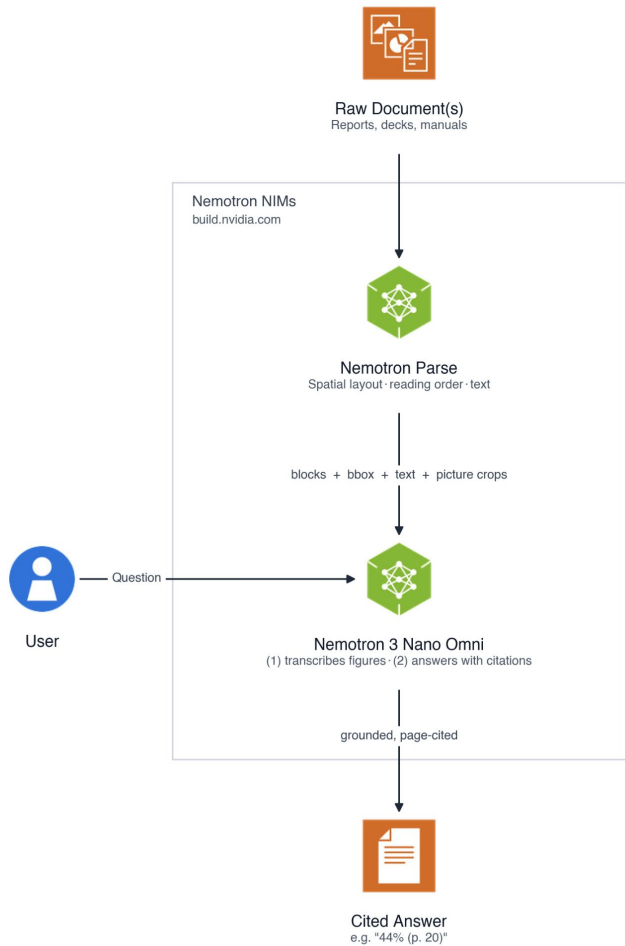
Scans the entire 116-page filing



Finds and extracts relevant evidence



Combines distant evidence to answer



For complex OCR it's best to pair with Nemotron Parse



All-Modality Document AI with Nemotron Parse + Nemotron 3 Nano Omni

You do not need a GPU to run this notebook. Every model call goes to NVIDIA's hosted catalog endpoint at `https://integrate.api.nvidia.com/v1` (the same backend that powers build.nvidia.com). All you need is a free `NVIDIA_API_KEY` — see Setup below for a 30-second walkthrough.

Want to self-host instead? Both models ship as NIMs on NGC -- see the **Deploy** tab on each model card for copy-paste vLLM / SGLang / TensorRT-LLM launch commands ([Parse](#) · [Deploy](#) · [Nano Omni](#) · [Deploy](#)) -- and as open weights (BF16 / FP8 / NVFP4) on Hugging Face. Tested hardware: H100, B200, DGX Spark, RTX Pro 6000, Jetson Thor.

This notebook builds a streamlined, **all-modality** document analysis pipeline by pairing two NVIDIA models. The star of the pipeline is **Nemotron 3 Nano Omni** (`nvidia/nemotron-3-nano-omni-30b-a3b-reasoning`) — a single multimodal reasoning model that both **reads every picture on the page** and **answers questions about the whole document**, keeping what the picture said and what the question needs inside one brain. Alongside it, **Nemotron Parse** (`nvidia/nemotron-parse`) provides the **spatial anchoring** that lets Nano Omni focus on one region at a time and cite the exact page and phrase its answer came from.

To make the case concretely, we run the pipeline against four pages picked from **three different public PDFs** — a Pew Research report, a social-media analytics deck, and a graduate-studies brochure — each chosen to stress a **different content modality** that a single VLM cannot handle alone:

https://github.com/NVIDIA-NeMo/Nemotron/blob/main/usage-cookbook/Nemotron-3-Nano-Omni/doc-intelligence-with-parse/doc_intelligence_cookbook.ipynb

Summary

Nemotron 3 Omni unifies text, vision, and audio in a single model

- Embeddings enable flexible multimodal workflows
- Two modes: descriptive and reasoning

Powers end-to-end pipelines for document intelligence, multimodal search, and assistants

- Multimodal prompts let the model attend to specific regions, layouts, or visual details
- Integrates with tools like Nemotron Parse and RAG to build production-ready multimodal apps

Resources

- [Nemotron 3 Nano Omni - Technical Report](#)
- [Nemotron Developer repository at GitHub](#)
- [Nemotron 3 Nano Omni 30B @ HuggingFace](#)
- [Nemotron Parse 1.2 @ HuggingFace](#)
- [Practical Computer Vision Bootcamp - GitHub Repository](#)

Grow Your Skills with NVIDIA Learning Paths

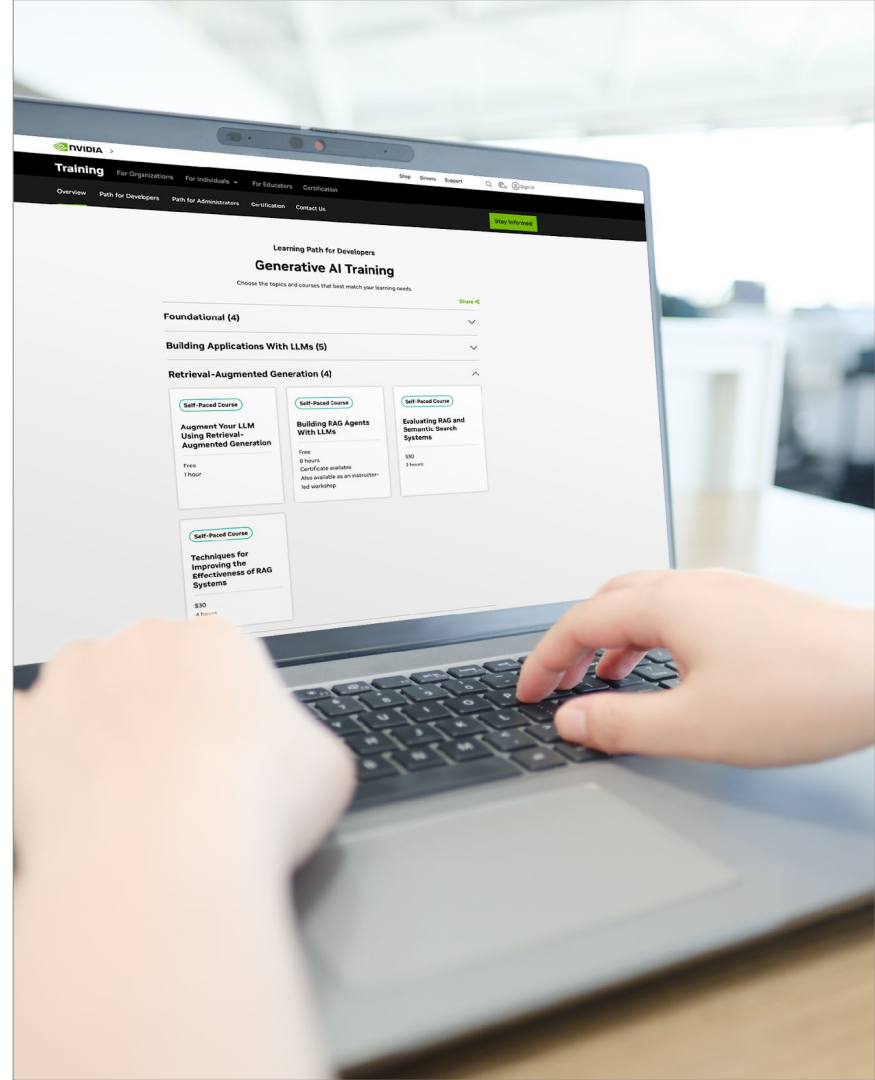
Build expertise in AI, accelerated computing, and graphics through guided NVIDIA learning paths

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- Follow structured, role-based skill development
- Learn through hands-on labs and real-world projects
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<https://nvda.ws/3ZXQYb7>



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Workshops October 20, 2026

Conference and Expo October 21–22



nvidia.com/en-eu/gtc

